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Valve with docking station for gastrointestinal balloon

Abstract

A gastrointestinal balloon system includes a gastrointestinal balloon and a valve located within the gastrointestinal balloon. The valve is configured to allow fluid flow into and out of the gastrointestinal balloon. The valve includes a string coupled to the valve, the string having a loop on a proximal end and a catheter having a lumen. The lumen is configured to receive a grasping tool. The grasping tool extends out a distal end of the catheter for grabbing the loop to align the catheter with the valve. The valve may be a series of self-sealing two-way valves, a check valve, or two-way valve. A method of using a valve in a gastrointestinal balloon is also described.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3726283	12/1972	Dye et al.	N/A	N/A
4102342	12/1977	Akiyama et al.	N/A	N/A
4116201	12/1977	Shah	N/A	N/A
4129145	12/1977	Wynn	N/A	N/A
4311146	12/1981	Wonder	N/A	N/A
4315509	12/1981	Smit	N/A	N/A
4364392	12/1981	Strother	606/195	A61B 17/0057
4416267	12/1982	Garren et al.	N/A	N/A
4447227	12/1983	Kotsanis	N/A	N/A
4471779	12/1983	Antoshkiw	604/907	A61B 17/12109
4485805	12/1983	Foster, Jr.	N/A	N/A
4598699	12/1985	Garren et al.	N/A	N/A
4694827	12/1986	Weiner et al.	N/A	N/A
4696288	12/1986	Kuzmak et al.	N/A	N/A
4738667	12/1987	Galloway	N/A	N/A
4739758	12/1987	Lai et al.	N/A	N/A
4766628	12/1987	Walker	N/A	N/A
4813934	12/1988	Engelson et al.	N/A	N/A
4878905	12/1988	Blass	N/A	N/A
4899747	12/1989	Garren et al.	N/A	N/A
4908011	12/1989	Jacobsen et al.	N/A	N/A
4925446	12/1989	Garay et al.	N/A	N/A
5002556	12/1990	Ishida et al.	N/A	N/A
5052998	12/1990	Zimmon	N/A	N/A
5084061	12/1991	Gau et al.	N/A	N/A
5129915	12/1991	Cantenys	N/A	N/A
5167239	12/1991	Cohen et al.	N/A	N/A
5234454	12/1992	Bangs	N/A	N/A
5259399	12/1992	Brown	N/A	N/A
5380282	12/1994	Burns	N/A	N/A
5433216	12/1994	Sugrue et al.	N/A	N/A

5458583	12/1994	McNeely et al.	N/A	N/A
5536274	12/1995	Neuss	N/A	N/A
5578048	12/1995	Pasqualucci et al.	N/A	N/A
5624395	12/1996	Mikhail et al.	N/A	N/A
5724994	12/1997	Simon et al.	N/A	N/A
5732715	12/1997	Jacobs et al.	N/A	N/A
5779672	12/1997	Dormandy, Jr.	604/99.04	A61B 17/12136
5827304	12/1997	Hart	N/A	N/A
5976106	12/1998	Verin et al.	N/A	N/A
5985307	12/1998	Hanson et al.	N/A	N/A
6050972	12/1999	Zadno-Azizi et al.	N/A	N/A
6127597	12/1999	Beyar et al.	N/A	N/A
6183461	12/2000	Matsuura et al.	N/A	N/A
6213936	12/2000	Nishioka et al.	N/A	N/A
6245040	12/2000	Inderbitzen et al.	N/A	N/A
6364686	12/2001	Gimbel	N/A	N/A
6364868	12/2001	Ikeguchi	N/A	N/A
6569173	12/2002	Blatter et al.	N/A	N/A
6616678	12/2002	Nishtala et al.	N/A	N/A
6652339	12/2002	Carmichael	N/A	N/A
6656194	12/2002	Gannoe et al.	N/A	N/A
6712831	12/2003	Kaplan et al.	N/A	N/A
6743198	12/2003	Tihon	N/A	N/A
6746460	12/2003	Gannoe et al.	N/A	N/A
7004176	12/2005	Lau	N/A	N/A
7033384	12/2005	Gannoe et al.	N/A	N/A
7056305	12/2005	Garza Alvarez	N/A	N/A
7087009	12/2005	Migachyov	N/A	N/A
7147665	12/2005	Bryan et al.	N/A	N/A
7244270	12/2006	Lesh	623/1.11	A61M 29/02
7331949	12/2007	Marisi	N/A	N/A
7347868	12/2007	Burnett et al.	N/A	N/A
7461688	12/2007	Huang	165/104.21	F28D 15/0266
7470251	12/2007	Shah	N/A	N/A
7628821	12/2008	Stack et al.	N/A	N/A
7632291	12/2008	Stephens	606/195	A61B 17/12113
7641688	12/2009	Lesh	623/23.72	A61M 29/02
7654950	12/2009	Yachia et al.	N/A	N/A
7753928	12/2009	de la Torre et al.	N/A	N/A
7883524	12/2010	Chen	N/A	N/A
8062207	12/2010	Gannoe et al.	N/A	N/A
8088103	12/2011	Teeslink et al.	N/A	N/A
8403952	12/2012	Brooks et al.	N/A	N/A
8430894	12/2012	Brooks et al.	N/A	N/A
8888739	12/2013	Gregory et al.	N/A	N/A
8939991	12/2014	Krolik et al.	N/A	N/A
9044571	12/2014	Pinchuk et al.	N/A	N/A

9173757	12/2014	Birk	N/A	N/A
9308361	12/2015	Muni et al.	N/A	N/A
9399121	12/2015	Goldfarb et al.	N/A	N/A
9504811	12/2015	Tilson et al.	N/A	N/A
9974680	12/2017	Brooks	N/A	N/A
10857019	12/2019	Brooks	N/A	N/A
10893966	12/2020	Brooks et al.	N/A	N/A
2001/0020150	12/2000	Ravo	N/A	N/A
2002/0045914	12/2001	Roberts et al.	N/A	N/A
2002/0055757	12/2001	Torre et al.	N/A	N/A
2002/0151923	12/2001	Holzer	N/A	N/A
2003/0158569	12/2002	Wazne	N/A	N/A
2003/0171768	12/2002	McGhan	N/A	N/A
2003/0191492	12/2002	Gellman et al.	N/A	N/A
2004/0030347	12/2003	Gannoe et al.	N/A	N/A
2004/0044357	12/2003	Gannoe et al.	N/A	N/A
2004/0059289	12/2003	Garza Alvarez	N/A	N/A
2004/0064893	12/2003	Sharp	N/A	N/A
2004/0088008	12/2003	Gannoe et al.	N/A	N/A
2004/0093091	12/2003	Gannoe et al.	N/A	N/A
2004/0167625	12/2003	Beyar et al.	N/A	N/A
2004/0193092	12/2003	Deal	N/A	N/A
2004/0225247	12/2003	Pugsley et al.	N/A	N/A
2004/0267378	12/2003	Gazi et al.	N/A	N/A
2005/0004430	12/2004	Lee et al.	N/A	N/A
2005/0033331	12/2004	Burnett et al.	N/A	N/A
2005/0033345	12/2004	DeLegge	N/A	N/A
2005/0055039	12/2004	Burnett et al.	N/A	N/A
2005/0070937	12/2004	Jambor et al.	N/A	N/A
2005/0085923	12/2004	Levine et al.	N/A	N/A
2005/0096750	12/2004	Kagan et al.	N/A	N/A
2005/0125020	12/2004	Meade et al.	N/A	N/A
2005/0192614	12/2004	Binmoeller	N/A	N/A
2005/0192629	12/2004	Saadat et al.	N/A	N/A
2005/0209674	12/2004	Kutscher et al.	N/A	N/A
2005/0216040	12/2004	Gertner et al.	N/A	N/A
2005/0228504	12/2004	Demarais	N/A	N/A
2005/0256460	12/2004	Rome	604/247	A61M 39/26
2005/0267361	12/2004	Younker et al.	N/A	N/A
2005/0267595	12/2004	Chen et al.	N/A	N/A
2005/0267596	12/2004	Chen et al.	N/A	N/A
2006/0020278	12/2005	Burnett et al.	N/A	N/A
2006/0058735	12/2005	Lesh	N/A	N/A
2006/0058892	12/2005	Lesh et al.	N/A	N/A
2006/0100479	12/2005	Yachia et al.	N/A	N/A
2006/0142731	12/2005	Brooks	N/A	N/A
2006/0190019	12/2005	Gannoe et al.	N/A	N/A
2006/0206064	12/2005	Kagan et al.	N/A	N/A
2006/0271088	12/2005	Alfrhan	N/A	N/A
2007/0083224	12/2006	Hively	N/A	N/A

2007/0088380	12/2006	Hirszowicz et al.	N/A	N/A
2007/0118168	12/2006	Lointier et al.	N/A	N/A
2007/0149994	12/2006	Sosnowski et al.	N/A	N/A
2007/0173881	12/2006	Birk et al.	N/A	N/A
2007/0239284	12/2006	Skerven et al.	N/A	N/A
2008/0051823	12/2007	Makower	606/192	A61B 17/1285
2008/0108939	12/2007	Moulton	604/43	A61M 39/045
2008/0147012	12/2007	Rome	N/A	N/A
2008/0172079	12/2007	Birk	606/192	A61F 5/0036
2008/0208135	12/2007	Annunziata	N/A	N/A
2009/0152379	12/2008	Harter	N/A	N/A
2009/0164028	12/2008	Chen	N/A	N/A
2009/0171382	12/2008	Dillon et al.	N/A	N/A
2009/0287231	12/2008	Brooks et al.	N/A	N/A
2010/0121371	12/2009	Brooks et al.	N/A	N/A
2010/0252464	12/2009	Belitz	N/A	N/A
2010/0274189	12/2009	Kurth et al.	N/A	N/A
2011/0092998	12/2010	Hirszowicz et al.	N/A	N/A
2011/0218563	12/2010	Brooks et al.	N/A	N/A
2012/0191123	12/2011	Brister et al.	N/A	N/A
2012/0296365	12/2011	Nguyen	N/A	N/A
2013/0035710	12/2012	Bouasaysy et al.	N/A	N/A
2013/0255807	12/2012	Bonzon et al.	N/A	N/A
2014/0188151	12/2013	Gaur et al.	N/A	N/A
2014/0249374	12/2013	Evans et al.	N/A	N/A
2016/0100946	12/2015	Fogarty et al.	N/A	N/A
2016/0193064	12/2015	Brister et al.	N/A	N/A
2016/0310306	12/2015	Brister et al.	N/A	N/A
2017/0007310	12/2016	Rajagopalan et al.	N/A	N/A
2017/0156909	12/2016	Brister et al.	N/A	N/A
2017/0325982	12/2016	Brooks et al.	N/A	N/A
2017/0354437	12/2016	Bacich	N/A	N/A
2018/0209553	12/2017	Weaver	N/A	N/A
2018/0221184	12/2017	Brooks et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1233387	12/1987	CA	N/A
2068715	12/1991	CA	N/A
2483335	12/2004	CA	N/A
104968304	12/2014	CN	N/A
204890270	12/2014	CN	N/A
3227585	12/1982	DE	N/A
B326061	12/1983	DE	N/A
3310234	12/1983	DE	N/A
3540936	12/1985	DE	N/A
246999	12/1986	EP	N/A

1342458	12/2004	EP	N/A
2004269	12/2015	EP	N/A
2090747	12/1981	GB	N/A
2139902	12/1983	GB	N/A
1235492	12/1991	IT	N/A
2008517677	12/2007	JP	N/A
1986006611	12/1985	WO	N/A
1987000034	12/1986	WO	N/A
1998056321	12/1997	WO	N/A
2001066166	12/2000	WO	N/A
2002040081	12/2001	WO	N/A
2003055420	12/2002	WO	N/A
2003095015	12/2002	WO	N/A
2004014237	12/2003	WO	N/A
2004089262	12/2003	WO	N/A
2004105622	12/2003	WO	N/A
2005009288	12/2004	WO	N/A
2005039457	12/2004	WO	N/A
2005094257	12/2004	WO	N/A
2005107641	12/2004	WO	N/A
2005094257	12/2005	WO	N/A
2006047708	12/2005	WO	N/A
2007110866	12/2006	WO	N/A
2008042819	12/2007	WO	N/A
2008132745	12/2007	WO	N/A
2006070361	12/2008	WO	N/A
2011041864	12/2010	WO	N/A
2011097636	12/2010	WO	N/A
2016/200612	12/2015	WO	N/A
2018146547	12/2017	WO	N/A

OTHER PUBLICATIONS

Andersen et al., “Randomised trial of endoscopic endoprosthesis versus operative bypass in malignant obstructive jaundice”, Gut, 30, 1132-1135 (1989). cited by applicant

Borowski A et al., “Minimally Invasive, Nonendoscopic Saphenectomy for Coronary Bypass Surgery”, J Card Surg 16(6): 484-6, (2001)—an abstract. cited by applicant

Cope, “Improved Anchoring of Nephrostomy Catheters: Loop Technique”, American J of Roetngenology 1980. cited by applicant

Dondelinger et al, “Percutaneous management of intraperitoneal, hepatic and other fluid collections”, Baillieres Clin Gastroenterol (1992). cited by applicant

Dumon, J.R. et al., “A new method of esophageal dilation using Savary-Gilliard Bougies,” Gastro Endosc 31:379-82, 1985. cited by applicant

Fleischer, D.E. et al., “A marked guidewire facilitates esophageal dilation,” Am J Gastro 84:359-61, 1989. cited by applicant

Gronval et al , “Ultrasound-guided Drainage of Fluid-containing Masses Using Angiographic Catheterization Techniques”, Am J Roetgenol (1977). cited by applicant

Kadokia, SC et al., “Esophageal dilation with polyvinyl bougies using a marked guidewire without the aid of fluoroscopy,” Am J Gastro 88:1381-86, 1993. cited by applicant

Lambiase, “Percutaneous Abscess and Fluid Drainage: A Critical Review”, Cardiovascular and interventional radiology 14: 143-157 (1991). cited by applicant

Parker et al. "Esophageal dilation with polyvinyl bougies, using a marked guidewire without the aid of fluoroscopy: an update", *Am J Gastroenterol.* (Sep. 1993); 88(9):1381-6—an abstract. cited by applicant

Shepherd et al, "Endoscopic biliary endoprosthesis in the palliation of malignant obstruction of the distal common bile duct: a randomized trial", *British journal of surgery*, vol. 75, pp. 1166-1168 (1988). cited by applicant

Speer et al., "Randomised Trial of Endoscopic Versus Percutaneous Stent Insertion in Malignant Obstructive Jaundice", *The Lancet* (1987). cited by applicant

Van sonnenberg et al., "Percutaneous Abscess Drainage: Current Concepts", *Radiology* (1991). cited by applicant

Werth, et al., "A safe and quick method for endoscopic retrieval of multiple gastric foreign bodies using a protective sheath," *Surg Gynecol Obstet* 171(5):419-20, 1990. cited by applicant

Translation of Office Action issued in related Chinese Patent Application No. 200580048692.5 dated Apr. 13, 2010. cited by applicant

Supplementary European Search Report issued in related European Application No. 07736138 dated Aug. 13, 2013. cited by applicant

European Search Report issued in related European Patent Application No. 11161677.7 dated Aug. 22, 2011. cited by applicant

Office Action issued in related Israel Patent Application No. 183649 dated Jan. 21, 2010. cited by applicant

Notification of Reasons for Refusal issued in Japanese Patent Application No. 2007-548959 dated Apr. 26, 2011. cited by applicant

Notification of Reasons for Refusal issued in related Japanese Patent Application No. 2009-502330 dated Feb. 21, 2012. cited by applicant

Notification of Reasons for Refusal issued in related Japanese Patent Application No. 2011-56619 dated Oct. 16, 2012. cited by applicant

International Search Report issued in International Patent Application No. PCT/IL05/01381 dated May 28, 2008. cited by applicant

International Search Report issued in International Patent Application No. PCT/IL07/00398 dated Sep. 22, 2008. cited by applicant

International Search Report issued in International Patent Application No. PCT/IL08/00579 dated Dec. 23, 2008. cited by applicant

International Search Report issued in International Patent Application No. PCT/IL 10/00833 dated Feb. 10, 2011. cited by applicant

International Search Report and Written Opinion issued in International Patent Application No. PCT/US2013/071730 dated Mar. 10, 2014. cited by applicant

Office Action issued in U.S. Appl. No. 11/132,855 dated Oct. 16, 2008 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/132,855 dated Jun. 11, 2009 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/132,855 dated Dec. 30, 2009 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/132,855 dated Jun. 25, 2010 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/132,855 dated Aug. 24, 2011 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/132,855 dated Mar. 27, 2012 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/718,795 dated Sep. 17, 2010 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/721,993 dated Feb. 27, 2012 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 12/598,110 dated Dec. 13, 2011 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 12/903,448 dated Sep. 18, 2012 [Available in IFW]. cited by applicant

U.S. Appl. No. 60/639,843, filed Dec. 27, 2004 [Available in IFW]. cited by applicant

U.S. Appl. No. 60/787,124, filed Mar. 28, 2006 [Available in IFW]. cited by applicant

U.S. Appl. No. 60/815,624 dated Jun. 21, 2006 [Available in IFW]. cited by applicant

Indian Examination Report issued in corresponding Indian Application No. 9024/DELNP/2008 dated Jan. 27, 2017. cited by applicant

U.S. Appl. No. 62/457,056, filed Feb. 9, 2017 [Available in IFW]. cited by applicant

International Search Report and Written Opinion issued in related International Application No. PCT/IB18/00173 dated Jul. 12, 2018. cited by applicant

Office Action issued in U.S. Appl. No. 12/598,110 dated Nov. 7, 2013 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/718,795 dated Jul. 8, 2011 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 11/718,795 dated Oct. 9, 2012 [Available in IFW]. cited by applicant

Office Action issued in U.S. Appl. No. 14/647,372 dated Oct. 2, 2017 [Available in IFW]. cited by applicant

Office Action issued in corresponding Indian Application No. 9024/DELNP/2008 dated Nov. 5, 2019. cited by applicant

International Search Report and Written Opinion issued in related International Patent Application No. PCT/US2019/067921 dated Mar. 10, 2020. cited by applicant

Office Action issued in related U.S. Appl. No. 15/893,387 dated Feb. 5, 2020 [Available in IFW]. cited by applicant

Notice of Allowance dated Nov. 26, 2012, directed to U.S. Appl. No. 11/721,993; 5 pages. cited by applicant

Notice of Allowance dated Feb. 28, 2013, directed to U.S. Appl. No. 11/718,795; 9 pages. cited by applicant

Notice of Allowance dated Feb. 21, 2018, directed to U.S. Appl. No. 14/647,372; 8 pages. cited by applicant

Corrected Notice of Allowability dated Mar. 1, 2018, directed to U.S. Appl. No. 14/647,372; 2 pages. cited by applicant

Non-Final Rejection dated Apr. 17, 2020, directed to U.S. Appl. No. 15/949,342; 12 pages. cited by applicant

Final Office Action dated May 12, 2020, directed to U.S. Appl. No. 15/893,387; 14 pages. cited by applicant

Advisory Action dated Jul. 27, 2020, directed to U.S. Appl. No. 15/893,387; 14 pages. cited by applicant

Notice of Allowance dated Aug. 13, 2020, directed to U.S. Appl. No. 15/949,342; 9 pages. cited by applicant

Notice of Allowance dated Sep. 17, 2020, directed to U.S. Appl. No. 15/893,387; 11 pages. cited by applicant

Corrected Notice of Allowability dated Oct. 9, 2020, directed to U.S. Appl. No. 15/893,387; 7 pages. cited by applicant

Corrected Notice of Allowance dated Dec. 18, 2020, directed to U.S. Appl. No. 15/893,387; 3 pages. cited by applicant

Extended European Search Report dated Dec. 1, 2020, directed to EP Application No. 18751369.2; 7 pages. cited by applicant
Notification of the First Office Action dated Mar. 3, 2021, directed to CN Application No. 201880019146.7; 15 pages. cited by applicant
Examination Report mailed Jul. 29, 2021, directed to IN Application No. 201917035922; 5 pages. cited by applicant
Notification of the First Office Action dated Dec. 12, 2022, directed to CN Application No. 201980084870.2; 17 pages. cited by applicant
Search Report dated Sep. 7, 2022, directed to BR Application No. 112019016422-7; 4 pages. cited by applicant
Office Action dated Jun. 22, 2022, directed to MX Application No. MX/a/2019/009497; 4 pages. cited by applicant
Partial Supplementary European Search Report dated Aug. 12, 2022, directed to EP Application No. 19898750.5; 11 pages. cited by applicant
Examination Report dated Oct. 18, 2022, directed to IN Patent Application No. 202117031230; 6 pages. cited by applicant
Notice of Allowance dated Mar. 31, 2023 directed to MX Application No. MX/a/2019/009497; 3 pages. cited by applicant
Communication pursuant to Article 94(3) EPC dated May 10, 2023 directed to EP Application No. 18751369.2; 5 pages. cited by applicant
Second Office Action dated Jun. 27, 2023 directed to CN Application No. 201980084870; 18 pages. cited by applicant
Official Communication dated Jul. 5, 2023 directed to BR Application No. BR112021011978-7; 4 pages. cited by applicant
Final Rejection dated Apr. 26, 2024, directed to U.S. Appl. No. 17/118,025; 12 pages. cited by applicant
Notice of Refusal dated Jul. 4, 2024, directed to Brazilian Patent Application No. 12 2023 001352-2; 3 pages. cited by applicant
Advisory Action dated Jul. 9, 2024, directed to U.S. Appl. No. 17/118,025; 3 pages. cited by applicant
Notice of Allowance dated Aug. 14, 2024, directed to U.S. Appl. No. 17/118,025; 9 pages. cited by applicant
Notice of Refusal dated Jan. 23, 2024, directed to BR Patent Application No. 11 2019 016422 7; 5 pages. cited by applicant
Office Action dated Jan. 24, 2024, directed to BR Patent Application No. 12 2023 001352 2; 6 pages. cited by applicant
Hearing Notice dated Mar. 21, 2024, directed to IN Patent Application No. 202117031230; 3 pages. cited by applicant
Notification of Third Office Action dated Oct. 27, 2023, directed to CN Patent Application No. 201980084870.2; 13 pages. cited by applicant
Non-Final Rejection dated Oct. 25, 2023, directed to U.S. Appl. No. 17/118,025; 9 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 62/784,106 filed Dec. 21, 2018, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD OF THE INVENTION

(1) The present invention relates generally to implantable medical devices and systems that include a check valve, and methods of using implantable medical devices and systems with a check valve. Specifically, the present invention relates to devices, systems, and methods for a check valve and a gastrointestinal balloon including a check valve. The present invention also relates generally to a check valve with docking station. The check valve with docking station may be in a remote location or a difficult to access locations. The present invention relates to valves for access to a remote location, such as, a gastrointestinal balloon located in a patient. The present invention relates to a series of valves with a docking station.

BACKGROUND OF THE INVENTION

(2) Morbid obesity remains an ever-growing problem in the U.S. Varying forms of gastric bypass surgery have developed and have improved over the last few decades. Recently, laparoscopic gastric bypass has emerged as a less invasive surgical option. However, bariatric surgery is fraught with morbidity of up to 20%, with a re-operation rate approaching 25% at 3-5 years post-operation. Bariatric surgery carries an operative mortality of 0.5%. Diet and pharmaceutical alternatives have not been very effective, with a high recidivism rate. Intragastric balloons in use may achieve weight loss and a drop in body mass index (BMI). However, it is desirable to be able to change the volume of the balloon while in the stomach of a patient, either to increase the balloon volume or decrease the balloon volume, as balloon effect wears off after several months and enlarging the balloon can rejuvenate the balloon effect, and some balloons may cause intolerance and require balloon volume decrease.

(3) There is a need to be able to change the volume of a gastrointestinal balloon while the balloon is in the stomach of a patient. In order to access the balloon valve to adjust balloon volume while the balloon is still in the stomach, there are two options. One option is to bring the valve out of the stomach to the mouth of a patient, where balloon volume may be manually adjusted. For example, some existing gastrointestinal balloon systems use a stretchable inflation tube that may stretch from a stomach to the mouth of the patient, or about 2.5-10 times its length, to facilitate inflation or adjustment of volume of the balloon. The inflation tube is partly inside and partly outside of the balloon. U.S. Patent Application Publication No. 2006/0142731 and U.S. Pat. No. 8,403,952, which are incorporated herein by reference, describe a floating anchor that may be used, for example, with the gastrointestinal balloon. However, further improvements may be made to limit side effects from implantable devices including, but not limited to, tissue trauma, difficulty finding the tube, and difficulty grasping the tube, some of which may result of components of the devices that are located on the outside of the gastrointestinal balloon.

(4) Another option is to keep the valve stationary in the balloon and access the valve itself while the valve resides in the stomach, without removing any of the components from the stomach. However, direct access to the balloon valve while the valve is within the stomach via an endoscope is extremely difficult and may not be feasible because the positioning of the endoscope relative to the balloon valve requires pinpoint precision with head-on engagement. This is a technically difficult maneuver for an endoscope engaging a spherical balloon in the wet environment of the stomach. For example, the balloon may easily move within the stomach while, or as a result of, attempting to engage the valve.

(5) Thus, there exists a need for improved gastrointestinal balloon devices, systems, and methods

for accessing a balloon valve while it is in the stomach in a way that provides a high success rate for engaging with the valve to adjust the balloon volume.

BRIEF SUMMARY OF THE INVENTION

(6) According to embodiments of the invention, a gastrointestinal balloon is immobilized within the stomach to improve the ease of engaging the valve. In particular, the balloon may be immobilized while at the same time allowing access to the valve at the correct angle by the inflation tool. A check valve is a type of valve that may be used for changing the volume of a gastrointestinal balloon. However, a standard check valve cannot be simultaneously immobilized and accessed with current check valve configurations and technology.

(7) Thus, novel devices, systems, and methods are provided herein for a check valve used in inflatable medical devices, including but not limited to gastrointestinal balloons that may be simultaneously immobilized and accessed while in a body cavity. After immobilization of the balloon and aligning of the inflation tool to the balloon valve, the balloon may be inflated or deflated.

(8) Additional features, advantages, and embodiments of the invention are set forth or apparent from consideration of the following detailed description and drawings. Moreover, it is to be understood that both the foregoing summary of the invention and the following detailed description are exemplary and intended to provide further explanation without limiting the scope of the invention as claimed.

(9) According to an embodiment of the present disclosure, a check valve for use in a gastrointestinal balloon may include a housing having an entry, the entry configured to allow fluid flow into and out of the gastrointestinal balloon; a plunger configured to open and close the entry; and an attachment extending from a proximal end of the housing. The attachment is configured to be grasped by a tool to permit aligning of a catheter with a proximal end of the housing.

(10) The check valve includes an open position and a closed position, wherein the check valve is configured to be moved to the open position in a stomach. The check valve is configured to be mobile in a stomach. The catheter is configured to impact the proximal end of the housing. The plunger further comprising a plunger head, a plunger arm, and a plunger tail. The plunger head includes an opening for the attachment to extend through and a seal for sealing the opening. The plunger arm is formed from a cylindrical body with two semi-circular arms extending between the cylindrical body and a distal end of the plunger head. The plunger tail is coupled to a distal end of the plunger arm. The plunger head, plunger arm, and plunger tail are integrally formed as a single unit. An o-ring seal is located between the plunger arm and the plunger tail. A block is coupled to the housing and coupled to the attachment. A spring is coupled between the block and a proximal end of the housing. An opening in the proximal end of the housing is configured to removably receive a distal end of the catheter.

(11) According to an embodiment of the present disclosure, a gastrointestinal balloon system, may include a gastrointestinal balloon; a valve located within the gastrointestinal balloon, the valve configured to allow fluid flow into and out of the gastrointestinal balloon; an attachment coupled to the valve, the attachment having a connector on a proximal end; and a shaft having a lumen, the lumen configured to receive a grasping tool. The grasping tool extends out a distal end of the shaft for grabbing the connector to align the shaft with the valve.

(12) The shaft is a catheter and the catheter has a tapered distal tip. The valve separates an interior of the gastrointestinal balloon with an exterior of the gastrointestinal balloon. The gastrointestinal balloon is configured to be inflated or deflated through the valve. The valve includes a proximal opening, a plunger, and an o-ring to seal an entry into the gastrointestinal balloon. The valve is a check valve. The attachment is a string.

(13) According to an embodiment of the present disclosure, a method of accessing a gastrointestinal balloon may include inserting a catheter through a channel of a gastroscope and into a stomach of a patient; inserting a grasping tool through a lumen of the catheter and extending

the grasping tool through a distal end of the catheter; grasping an attachment with the grasping tool, wherein the attachment is coupled to a check valve; moving the check valve into alignment with the catheter; moving a plunger in the check valve such that the check valve is moved from a closed position to an open position; and inflating or deflating the gastrointestinal balloon through the check valve.

(14) The method includes removably impacting the catheter into the check valve. The method includes impacting the catheter into the check valve comprises impacting a distal end of the catheter with a proximal opening of a housing of the check valve. The method includes moving the plunger in the check valve comprises distally moving the plunger in the check valve. The method includes releasing the attachment from the grasping tool and removing the grasping tool from the catheter prior to inflating or deflating the gastrointestinal balloon. The distal end of the catheter is tapered.

(15) According to an embodiment of the present disclosure, a check valve system may include a check valve having a housing, the housing having a distal opening and a proximal opening, the distal opening configured to be sealed; an elongated member coupled to the housing, the elongated member having a proximal end with a connector; a tool configured to grasp the connector; and an impacting member configured to engage a proximal end of the housing. The tool is inserted through the impacting member to grasp the connector to align and immobilize the check valve with the impacting member. The tool is subsequently pulled to engage the impacting member into the proximal end of the housing to open the distal opening of the check valve. The check valve is configured for installation in a stomach, in a pipeline, in a plumbing line, or in a body cavity. A spring is configured to bias the check valve to a closed position.

(16) According to an embodiment of the present disclosure, a two-way valve may include a body configured to allow two-way flow. The body may have a docking end configured to receive a tool for positioning the body in a first position; and a port end configured to be disposed in a cavity. The valve has the first position and a second position. The docking end further includes an attachment to align the tool with the docking end.

(17) The first position is an open configuration and the second position is a closed configuration, the valve biased to the second position. The two-way valve is a check valve. The attachment is a strap. The docking end comprises an opening in a proximal end of the body. The opening is configured to receive the tool for releasably impacting the plunger.

(18) According to an embodiment, a valve for a gastrointestinal balloon, the valve comprising: a first valve formed of a self-sealing material; and an attachment extending from an interior of the gastrointestinal balloon through the first valve to an exterior of the gastrointestinal balloon.

(19) According to an embodiment, the valve further comprising a second valve located in series with the first valve, the second valve formed of a self-sealing material, wherein the attachment extends through the second valve.

(20) According to an embodiment, the valve further comprising a first spacer separating the first valve from the interior of the gastrointestinal balloon, the attachment extending through the first spacer.

(21) According to an embodiment, the valve further comprising a second spacer located in series and between the first valve and the second valve.

(22) According to an embodiment, wherein a first opening through the first valve and a second opening through the second valve are aligned.

(23) According to an embodiment, wherein the first opening and the second opening each have a first state preventing fluid flow therethrough and a second state allowing passage of a tool.

(24) According to an embodiment, wherein, in the first state, the first opening and the second opening are closed around the attachment.

(25) According to an embodiment, wherein the self-sealing material is silicone.

(26) According to an embodiment, wherein the first valve is formed as a cylindrical disc with an

opening therethrough, the opening configured to receive the attachment.

(27) According to an embodiment, further comprising an open position and a closed position, wherein the valve is configured to move to the open position in a stomach.

(28) According to an embodiment, wherein the open position permits passage of a tool and the closed position prohibits flow of fluid.

(29) According to an embodiment, further comprising a block configured to prevent removal of the attachment from the valve.

(30) According to an embodiment, the attachment extends through a center of the first valve.

(31) According to an embodiment, a gastrointestinal balloon system may include a self-sealing valve; an attachment extending through the self-sealing valve; and a tool configured to track over the attachment and pass through the self-sealing valve.

(32) According to an embodiment, wherein the tool is configured to centrally track over the attachment such that a central axis of the tool and the attachment are substantially coaxial.

(33) According to an embodiment, the self-sealing valve comprises a first self-sealing valve and a second self-sealing valve located in series.

(34) According to an embodiment, the self-sealing valve has an open position and a closed position, wherein, in the open position, the tool and the attachment pass through an opening of the self-sealing valve, and wherein, in the closed position, only the attachment passes through the opening of the self-sealing valve.

(35) According to an embodiment, a method of accessing a gastrointestinal balloon, the method comprising: grasping an attachment extending through a self-sealing valve; tracking a tool over the attachment and toward the self-sealing valve; pushing the tool through an opening in the self-sealing valve, still tracking over the attachment, thus opening the self-sealing valve; and inflating or deflating the gastrointestinal balloon through the tool.

(36) According to an embodiment, wherein tracking the tool over the attachment comprises passing the attachment through a lumen of the tool.

(37) According to an embodiment, wherein pushing the tool through an opening in the self-sealing valve comprises expanding the opening by entering a distal end of the tool into the opening.

(38) According to an embodiment, wherein the self-sealing valve is a series of two self-sealing valves formed of silicone.

(39) According to an embodiment, further comprising removing the tool from the self-sealing valve thus closing the self-sealing valve.

(40) According to an embodiment, a system for accessing an interior of a gastrointestinal balloon may include a gastrointestinal balloon; two contiguous valves; and a string running through a center of each of the two contiguous valves and exiting the gastrointestinal balloon, wherein the string is configured to be grasped by a user to track a tool over the string and directly through the two contiguous valves and into the interior of the gastrointestinal balloon.

(41) According to an embodiment, the two contiguous valves comprise a first valve located in series with a second valve, a first opening of the first valve and a second opening of the second valve being coaxially aligned.

(42) According to an embodiment, the two contiguous valves abut and touch each other.

(43) According to an embodiment, the system may include a spacer between the two contiguous valves.

(44) According to an embodiment, the system may include a tool configured to centrally track over the string and pass through the two contiguous valves.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate preferred embodiments of the invention and together with the detailed description serve to explain the principles of the invention. In the drawings:
- (2) FIG. 1 shows a standard check valve attached to a balloon, with the valve in the closed position.
- (3) FIG. 2 shows the standard check valve of FIG. 1, with the valve in the open position.
- (4) FIG. 3 shows a check valve in a gastrointestinal balloon according to an embodiment of the present invention.
- (5) FIG. 4 shows a close-up view of the check valve of FIG. 3, with the valve in the closed position.
- (6) FIG. 5 shows a close-up view of the check valve of FIG. 3 from a perspective rotated 90° relative to the view shown in FIG. 4.
- (7) FIG. 6 shows a close-up of the check valve of FIG. 3 in an open position according to an embodiment of the present invention.
- (8) FIG. 7 shows a catheter to be used in conjunction with the balloon and check valve and a step in a method of operating the check valve of FIG. 3, according to an embodiment of the present invention.
- (9) FIG. 8 shows a portion of the catheter, according to an embodiment of the present invention.
- (10) FIG. 9 shows a check valve in operation with a catheter and a grasping tool, according to an embodiment of the present invention.
- (11) FIG. 10A shows an exemplary grasping tool in a closed position, according to an embodiment of the present invention.
- (12) FIG. 10B shows an exemplary grasping tool in an open position, according to an embodiment of the present invention.
- (13) FIG. 11 shows a biluminal connector, according to an embodiment of the present invention.
- (14) FIG. 12 shows the biluminal connector of FIG. 11 with a coupling sleeve, according to an embodiment of the present disclosure.
- (15) FIG. 13 shows the biluminal connector and coupling sleeve of FIG. 12 coupled to a gastroscope, according to an embodiment of the present disclosure.
- (16) FIGS. 14A-14E show cross-sectional views of the biluminal connector of FIG. 11 taken along various axes of FIG. 11, according to an embodiment of the present disclosure.
- (17) FIG. 15 shows an alternative method for grasping the string, according to an embodiment of the present disclosure.
- (18) FIG. 16 shows a check valve in a gastrointestinal balloon according to another embodiment of the present invention.
- (19) FIG. 17 shows a close-up view of the check valve of FIG. 16, with the valve in the closed position.
- (20) FIG. 18 shows a close-up view of the check valve of FIG. 16 from a perspective rotated 90° relative to the view shown in FIG. 17.
- (21) FIG. 19 shows a close-up of the check valve of FIG. 16 in an open position according to an embodiment of the present invention.
- (22) FIG. 20 shows a catheter to be used in conjunction with the balloon and check valve of FIG. 16 and a step in a method of operating the check valve, according to an embodiment of the present invention.
- (23) FIG. 21 shows a check valve in operation with a catheter and a grasping tool, according to an embodiment of the present invention.
- (24) FIG. 22 shows a check valve in an open position with fluid flowing therethrough, according to an embodiment of the present invention.

- (25) FIG. 23 shows the check valve of FIG. 22 in the open position, rotated 90 degrees from the view of FIG. 22, according to an embodiment of the present invention.
- (26) FIG. 24 shows the check valve of FIG. 22 in the closed position, with fluid isolated on either side of the valve, according to an embodiment of the invention.
- (27) FIG. 25 shows a valve attached to a balloon, according to an embodiment of the present invention.
- (28) FIG. 26 shows a cross-sectional view of the valve of FIG. 25 taken along the section line 26-26.
- (29) FIG. 27 shows an exploded view of the valve of FIG. 25.
- (30) FIG. 28 shows an exploded view of the valve of FIG. 25.
- (31) FIG. 29 shows a valve attached to a balloon, according to an embodiment of the present invention.
- (32) FIG. 30 shows a cross-sectional view of the valve of FIG. 29 taken along the section line 29-29.

DETAILED DESCRIPTION OF THE INVENTION

(33) Devices, systems and methods are described herein with reference to the figures and in the context of medical devices, in particular a gastric balloon. In one aspect, a check valve with docking station can be provided to improve the ability to control the check valve for purposes of adjusting a gastric balloon (such as inflating and deflating the balloon). The check valve with docking station can improve the user's ability to change the volume of a gastrointestinal balloon while the balloon is in the stomach of a patient. The check valve with docking station can be used in other environments in order to access and adjust the valve in hard to reach areas where the devices may or may not be stabilized. The check valve docking features can be implement in other medical devices as well as non-medical devices and used as described herein in accordance with the principles of the invention. The check valve may include a plunger, string, spring, and valve housing. The string may be coupled to the housing such that when grasped and pulled, it aligns an opening of the valve housing with an impacting member for operating the opening and closing of the valve. Thus, the valve may be controlled from a remote location to allow fluid to flow into or out of the valve, as will be appreciated from the disclosure to follow.

(34) A standard check valve 10, as shown in FIGS. 1 and 2 (in the closed and opened positions, respectively), cannot be simultaneously immobilized and accessed with current check valve configurations and technology. FIGS. 1 and 2 depict a check valve 10 which separates the exterior 12 of the balloon from the interior or cavity 14 of the balloon. The check valve 10 may include a housing 16, plunger head 18, plunger arm 20, and a plunger tail 22. In FIG. 1, a spring 24, at a full extension length $L_{sub.E}$, keeps the check valve 10 in a closed position by pushing plunger head 18 away (to the left in the view shown in FIG. 1) from the mouth or entry 26 into the balloon cavity 14. The plunger head 18 in turn pulls the plunger arm 20 and the plunger tail 22 to the left and closes the entry 26 into the balloon cavity 14 by pulling an o-ring 28 to the left. In FIG. 2, The plunger head 18 is pushed toward (to the right in the view shown in FIG. 2) the entry 26 which contracts the spring 24 to a length $L_{sub.C}$, thereby allowing the plunger head 18 to move to the right (FIG. 2) towards the balloon cavity entry 26. The plunger head 18 then pushes the plunger arm 20 and the plunger tail 22 to the right which pushes the o-ring 28 to the right and opens for the bi-directional flow of fluid (arrows F) into and out of the balloon cavity 14.

(35) Engaging the check valve 10 in the balloon while the check valve 10 sits in the body cavity (e.g. the stomach) requires some method of immobilization and proper alignment of a tool on the check valve 10 of the balloon, followed by applying force to push the spring 24 towards the balloon cavity 14 (thereby opening flow F to the balloon cavity 14). Therefore, the present invention provides a novel modification to check valve technology that may accomplish these goals. The novel check valve has utility not only in the intragastric adjustable balloon, but also in other inflatable medical devices in any body cavity or check valves for industrial and non-medical

use, such as in plumbing. Therefore, references to “gastrointestinal balloons” or “stomach” are for example only, and are not intended to limit the scope of the invention.

(36) FIG. 3 shows a valve **100** according to an embodiment of the invention. The valve **100** may be a check valve, two-way valve, two directional flow valve, or other valve having an opening configured to be selectively opened and closed in accordance with the principle of the invention. For ease of description, the valve **100** is referred to herein as a check valve **100**. The check valve **100** may include a docking station to allow for alignment and engagement of an impacting member to open and close the check valve **100**. The check valve **100** is shown within a balloon **300** or integral with a sidewall **302** of the balloon **300**. The balloon **300** may be an adjustable balloon located in a patient's stomach. Alternatively, the check valve **100** may be located in a flow line of a plumbing or other hydraulic system. The balloon **300** may have an interior cavity **114** separated from an exterior space **112** (e.g. the stomach environment). The check valve **100** may separate the interior cavity **114** of the balloon **300** from the exterior space **112** of the balloon **300**. The check valve **100** may be provided with an attachment **200**, such as an elongated member **200**, such as a string **200** that may have a connector **202**. The connector **202** may be an “easy grasp” loop **202** that exits into the exterior space **112** of the balloon **300**. The string **200** with easy grasp loop **202** may be formed of nylon. The attachment **200** may be an elongated member, a bungee, a stretchy member, a string, a cable, a cord, other elongated device, or anything in accordance with the principles of invention capable of coupling to the check valve **100** and being grasped by a tool. For ease of description, the attachment **200** is described as a string **200** and the connector **202** is described as a loop **202**. The string **200** may be a single string **200** having opposing distal ends, each coupled to a block **130** (FIG. 4) of the housing of the check valve **100** with the easy grasp loop **202** formed therebetween.

(37) Moving (e.g. by pulling or pushing the impacting member and/or tool) the loop **202** into an impacting member **400** (FIG. 7), such as a catheter, may immobilize the balloon **300** and may guide the inflation/pushing catheter (not depicted) to the target (i.e., the check valve **100**), as will be described to follow. The string **200** may have a proximal end having a loop (such as easy grasp loop **202**) configured to be grasped by a tool. The easy grasp loop **202** may be pulled into a catheter by a grasping tool dispatched through an endoscope, as will be described in more detail. As may be appreciated, the balloon **300** may be located in a patient's stomach. The balloon **300** (including check valve **100**) may be floating or otherwise mobile within the stomach as the balloon is not secured to any wall therein. As previously mentioned, inflation and deflation of the balloon **300** may be difficult due to the mobile nature of the balloon **300**. In conventional applications, the balloon **300** is removed from the stomach. With the present check valve **100**, attachment **200**, and docking station, the balloon **300** is capable of being inflated or deflated while the balloon **300** remains in the patient's stomach. That is, as will be described below, the check valve **100** may be immobilized and aligned with a tool, the tool configured to open the valve and allow flow into or out of the balloon **300** as necessary. The fluid allowed to flow through the check valve **100** may be liquid, air, inflation fluid, or other fluid in accordance with principles of the invention.

(38) FIG. 4 depicts a close-up view of the check valve **100** with the balloon **300** omitted for clarity. The check valve **100** in FIG. 4 is depicted in a closed position. The check valve **100** may be biased in the closed position by a spring **124**. In this manner, once inflated or deflated to the appropriate level, the check valve **100** may retain the balloon **300** (not depicted) in the desired state. The check valve **100** may include a housing **116**, a plunger head **118**, a plunger arm **120**, a plunger tail **122**, the spring **124**, an o-ring **128**, and a block **130**. The housing **116** may be formed as a single piece or may be formed as a first housing portion **116a** and a second housing portion **116b** coupled together. The entry **126** may be formed integral with an interior surface of the housing **116** or may be a separate piece secured (e.g. with threads or fasteners) to an interior surface of the housing **116**. The block **130** may be built into, integrally formed with, or otherwise secured to the housing **116**. Thus, the block **130** may be a stationary, non-moving component against which the spring **124** may be

compressed. Alternatively, the block **130** may be permitted to move a small distance until engagement with an interior surface of the housing **116**. As may be appreciated, the spring **124** may be compressed between the surface **130a** of the block **130** and a distal surface of the plunger head **118**.

(39) In the check valve **100**, the plunger head **118**, plunger arm **120**, and plunger tail **122** may all be connected as one unit, similar to the check valve **10** of FIG. **1**. Alternatively, the plunger head **118**, plunger arm **120**, and plunger tail **122** may be formed integrally. The plunger arm **120** may be formed as a solid piece as is the plunger arm **20** of FIG. **1**. Alternatively, the plunger arm **120** may be split in the middle, leaving space for other parts, such as the string **200**, to be connected to the block **130** within the housing **116** of the check valve **100**, as will be described in relation to FIG. **5**.

(40) With continued reference to FIG. **4**, in the closed position, the spring **124** may be extended to a length $L_{sub.E}$ pushing the plunger head **118** to the left in FIG. **4**. In the closed position, the plunger tail **122** may hold the o-ring **128** in a sealing position with the entry **126** to the interior cavity (**114** in FIG. **3**) of the balloon **300** (FIG. **3**). The closed position of FIG. **4** may be the normal position of the check valve **100**. That is, the spring **124** may bias check valve **100** to a normally closed position. It may be appreciated that the plunger **118**, **120**, **122** may slide with respect to the block **130**. That is, the block **130** may extend through an interior of the plunger arm **120** (FIG. **5**) and the plunger may move with respect to the block **130** to compress the spring **124** against the block **130** and extend the o-ring **128** off of the entry **126** allowing flow therethrough.

(41) Referring now to FIG. **5**, a close-up view of the check valve **100** of FIG. **4** is shown, rotated 90° from the perspective of FIG. **4**. The plunger head **118**, plunger arm **120**, and plunger tail **122** of the check valve **100** may be formed as a single, integral piece or may be formed as separate parts coupled or secured together. The plunger head **118** may be generally cylindrical with a hole (not shown) through which the string **200** may pass. The plunger head **118** may be disc shaped. The hole may be located in the center of the plunger head **118** or may be located anywhere through plunger head **118** to facilitate connection of the string **200** to the block **130**. The plunger head **118** may further include a silicone seal **134** for sealing the hole and the string **200** within. The silicone seal may prevent fluid from passing through the hole having the string **200**. The plunger head **118** may include a cut out or groove **136** for coupling to the plunger arm **120**. Alternatively, the plunger head **118** and the plunger arm **120** may be formed as an integral piece.

(42) With continued reference to FIG. **5**, the plunger arm **120** may be cylindrical. The plunger arm **120** may be split in the middle such that a cavity **132** is located between opposing sides **120a** and **120b** of the plunger arm **120**. At the proximal end, the plunger arm sides **120a**, **120b** may be coupled or integral with the plunger head **118**. At the distal end, the plunger arm **120** main (cylindrical) body may be coupled to or integral with a neck (not shown) that is in turn coupled or integral with the plunger tail **122**. The neck may have a reduced diameter from the plunger arm **120** and/or plunger tail **122** such that the o-ring **128** may be fitted around the neck. The string **200** may have two ends **200a**, **200b** which extend through the plunger head **118** and through the cavity **132** of the plunger arm **120**. The two ends **200a** and **200b** may then be coupled, secured, or attached to the block **130**. Alternatively, the string **200** may be one or multiple strings with knots, loops, or components that may allow for grasping by a tool. As previously described, the plunger, including plunger arm **120**, may be allowed to move with respect to the block **130** and the housing **116**.

(43) Referring now to FIG. **6**, the check valve **100** is depicted in an open position, with the balloon **300** (FIG. **3**) removed for clarity. In the open position, the plunger head **118** may be pushed to the right in FIG. **6** within the housing **116**. Pushing or moving the plunger head **118** may shorten the spring **124** to the contracted length $L_{sub.c}$. As may be appreciated, the spring **124** may be secured between a proximal surface **118a** of the plunger head **118** and a distal surface **130a** of the block **130**. In this manner, the block **130** may operate as a stopper beyond which the spring **124** cannot pass. As the plunger head **118** is moved or pushed to the right, plunger arms **120** are correspondingly moved or pushed to the right due to the integral or coupled nature of the

components. The plunger arms **120** move the neck (not depicted) and plunger tail **122** into the interior cavity **114** (FIG. 3) of the balloon and out of the housing **116**. As the plunger tail **122** is moved out of the housing **116**, the o-ring **128** is moved away from the entry **126**, thus unseating the o-ring **128** from the entry **126** and allowing flow into and out of the interior cavity **114** of the balloon **300**.

(44) FIG. 7 shows an embodiment of an impacting member as a catheter **400**, which may, for example, have a 2-3.5 mm outer diameter with a wall thickness of 0.1 to 0.6 mm. The catheter **400** may be made of a semi-flexible material that allows the catheter to assume curves up to 90 degrees, yet at the same time maintain rigidity. Materials such as Teflon, low friction synthetic polymers, polymers coated with materials that produce a non-friction surface such as parylene or Teflon, and other material types may be employed. The catheter **400** may have a lumen with a diameter of 0.8 to 3.3 mm. The catheter **400** may be used in conjunction with the check valve **100**. The catheter **400** may be made of any suitable biocompatible material or material used for medical devices. The catheter **400** may have a tip **402**, but is not limited to the shape shown in FIG. 7. The tip **402** may be tapered, a luer, cone shaped, straight, or rounded tip. The tip **402** may facilitate entry of the catheter **400** into an opening **116c** of the housing portion **116b**. The opening **116c** may be a proximal opening of the housing **116** and the entry **126** may be a distal opening of the housing **116**. The catheter **400** may have a central lumen **404** through which a grasping tool **500** may extend. The grasping tool **500** may grab or secure the easy grasp loop **202** (FIG. 3) of the string **200**, as will be described to follow. The tip **402** may encounter the plunger head **118** during operation of the check valve **100** to move the plunger head **118** and thus open the check valve **100**. As may be appreciated, the proximal end of the housing portion **116b**, opening **116c**, the plunger head **118**, and the string **200** may be considered a docking station for the check valve **100**. That is, the catheter **400** is aligned with and impacted with these components such that alignment and operation of the check valve **100** are permitted. In this regard, the components act as a docking station for the catheter **400** and tool **500**.

(45) Referring to FIG. 8, a proximal part **406** of the catheter **400** may be Y-shaped with a homeostatic valve **408** at each arm that may be tightened or loosened to regulate flow and thereby prevent backflow outside of the catheter **400**. The proximal part **406** may include an inflation and/or deflation site **410** for inflating and/or deflating the balloon **300**. The proximal part **406** may have a handle with mechanical control that closes and opens the hook **502** (FIG. 10) of grasping tool **500**, or a handle that closes and opens the grasper of any type of grasping device.

(46) Referring to FIG. 9, during operation, an endoscope is extended into a patient's stomach. A catheter **400** with a grasping tool **500** are inserted therethrough. The grasping tool **500**, such as grasping forceps used in endoscopic procedures may be inserted into the lumen **404** of the catheter **400** through the homeostatic valve **408** (FIG. 8). During the procedure, the balloon, check valve, catheter, and grasping tool may all be located within the patient's stomach. The forceps or similar grasping tool **500** may grasp the string **200** by the easy grasp loop **202** and pull the check valve **100** toward the catheter **400**. At the same time, the catheter **400** may be advanced toward the check valve **100**. This may easily guide the grasping tool **500** to the target, the opening **116c**, of the check valve **100**. The catheter **400** then enters the opening **116c** of the housing **116** of the check valve **100**. Such alignment of the catheter **400** with the check valve **100** allows for immobilization of the check valve **100**.

(47) While fixating the catheter **400** and gastroscope with one hand, the grasping tool **500** (now grasping string **200**) within the catheter **400** is pulled back with the other hand. This motion causes the tip **402** to releasably engage the plunger head **118** of the check valve **100**. Continued pulling of the string **200** pushes the catheter tip **402** into releasably impacting the housing **116** and thereby causing the plunger head **118** to move to the right (as shown in the figures) or towards the interior cavity **114** of the balloon **300**. This movement causes the plunger arm **120** and the plunger tail **122** to move to the right which thereby unseats the o-ring **128** and opens the check valve **100**. At the

same time, the tip **402** of the catheter **400** impacts into the housing **116**. Alternatively, the catheter **400** may be straight, and the opening **116c** of the housing may be cone shaped resulting in the same impaction effect. The catheter **400** may impact into the housing **116**, which maintains the check valve **100** in an open position. The distal movement of the plunger head **118**, the plunger arm **120**, and the plunger tail **122**, the check valve **100** opens to bi-directional flow by separating the o-ring **128** from contacting the entry **126**. Once the catheter **400** is impacted and the check valve **100** is open, the string **200** may be released by the grasping tool **500** and the grasping tool **500** may be pulled out of the catheter **400**. This may open more space in the lumen **404** and allow enhanced flow of fluid. Alternatively, the grasping tool **500** may be left in place and flow may continue at a slower pace.

(48) After inflation or deflation is completed, the catheter **400** may be removed from the docking station or entry to the check valve **100**. The tool **500**, catheter **400**, and endoscope may be removed from the patient's stomach and from the patient.

(49) The force of the catheter **400** on the check valve **100** to impact and disengage the check valve **100** may be between 0.8 kg and 2 kg. The forces to contract and relax the spring **124** may be less than the forces that impact the valve by 25% to 40%. If the spring forces are greater than the impaction forces then the spring **124** may disengage the catheter **400** from the check valve. **100**

(50) With continued reference to FIGS. **3-9**, and described in another manner, during operation, the grasping tool **500** exits the endoscope and grasps the easy grasp loop **202** of the string **200**. The grasping tool **500** may be pulled while the tip **402** of the catheter **400** has entered into the check valve **100** of the balloon **300**. The catheter **400** may enter with ease, until the tip **402** broadens and impacts the plunger head **118** inside the housing **116** of the check valve **100**. The grasping tool **500** may be pulled hard to impact the catheter **400** into the check valve **100**. If the catheter **400** is not impacted into the check valve **100**, the check valve **100** may close prematurely. As the catheter **400** impacts inside the housing **116**, the catheter **400** pushes the plunger head **118** toward the entry **126** (compressing the spring **124**). This in turn pushes the plunger arm **120** and the plunger tail **122** forward, unseating the o-ring **128** from the entry **126**, and opens the check valve **100** to bi-directional flow. At this point the grasping tool **500** may release the string **200** (since the cone-tipped catheter **400** is impacted inside the housing **116**). The grasping tool **500** may then be pulled out of the catheter **400**. At this time, fluid may be admitted from the inflation site **410** (or removed through the deflation site **410**), though the lumen **404** of the catheter **400**, through the entry **126** of the check valve **100** and into the interior cavity **114** of the balloon **300**. With reference to FIG. **6**, the flow may enter the opening **116c** of the housing **116**, flow around the plunger head **118**, around and through the plunger arms **120**, through the entry **126**, and past the o-ring **128** and plunger tail **122**. It may be appreciated that during deflation, flow may follow the same path in the reverse direction. Once inflation/deflation is completed, the catheter **400** is pulled out of the check valve **100** and the spring **124** expands to its preferred position, which brings the plunger head **118**, plunger arm **120**, and plunger tail **122** back to the closed position (FIG. **5**) with the o-ring **128** sealed in the entry **126** and flow of fluid prevented.

(51) Although the catheter **400** is described as impacting the check valve **100**, the catheter may be forced into the valve in other manners. For example, the catheter **400** may be secured to the check valve opening **116c** with a turning motion. That is, the catheter **400** may have a tip **402** which may be screwed or threaded into the valve opening **116c**. One may appreciate that the inner surface of opening **116c** and the outer surface of the catheter tip **402** may have complementary threads or other surfaces to allow the rotational and longitudinal movement.

(52) FIGS. **10A** and **10B** depict the grasping tool **500** in an open (FIG. **10A**) and closed (FIG. **10B**) position. In the open position, the grasping tool **500** may have a hinged arm or hook **502** (or other connection type) that may align with the easy grasp loop **202** (FIG. **3**) of the string **200**. Once properly aligned, the hinged arm or hook **502** may be actuated to close thus enclosing the easy grasp loop **202** and securing the grasping tool **500** to the check valve **100** via the string **200**. Once

grasped, the grasping tool **500** may be pulled to pull the string **200** and thus the check valve **100** and balloon **300** into alignment with the catheter **400**. After proper alignment, the aforementioned impacting and opening of the check valve **100** may be performed. The grasping tool **500** may then be opened to release the check valve **100** (after impaction and/or opening). The grasping tool **500** may have a curved end that is hinged and can be opened to a straight position. In the closed position (FIG. **10B**), the length of the hinged arm or hook **502** is within 0.5 to 2 mm of the inner diameter of the catheter. The hinged arm or hook **502** may be moved into a position that can grasp (FIG. **10A**) or to a position that does not grasp (FIG. **10B**). The grasping tool **500** may alternatively be other types of grasping tools, such as a rat-tooth grasper or hook grasper.

(53) Thus, as may be appreciated from the foregoing disclosure, during a procedure to inflate and/or deflate the balloon, a catheter may be inserted into the channel of a standard flexible gastroscope, as practiced routinely by those skilled in the art of gastrointestinal endoscopy. As previously mentioned, the check valve and method of aligning and opening the check valve may be utilized in other body cavities or for industrial, plumbing, pipelines, wellbores, or other non-medical uses. For example, a check valve may be located in a pipeline. The pipeline may be inaccessible from the outer surface (e.g. a buried pipeline). The check valve may be biased to a normally closed position (as previously described) or biased to a normally open position. When it is desired to open or close the check valve, a grasping tool, similar to the grasping tool **500** but sized and dimensioned to fit in a pipeline instead of a catheter, may be extended into the pipeline inside another device (e.g. inside a tubular snake). Once inside the pipeline, the grasping tool may grasp a loop on a string coupled to the pipeline and align an impacting device (similar to the impacting catheter **400**) with an opening of the check valve. The grasping tool may pull the impacting device into alignment and impact the check valve in a manner similar to that previously described to effectuate opening and/or closing of the check valve.

(54) In an alternative embodiment shown in FIGS. **11-14**, the flow of inflation/deflation fluid may be separated from the lumen having the grasping tool **500**. There may be a need to separate the grasping tool **500** from the flow of inflation fluid for fear of gastric fluid contamination. This may be accomplished using a biluminal connector **600**. In this embodiment, the catheter **400** may be replaced with the biluminal connector **600** having a proximal end **602** which sits in a distal end of a channel of a gastroscope **800** (FIG. **13**). The biluminal connector **600** may have a distal end **604** which impacts the check valve **100** (not depicted). The distal end **604** may have a tip **606** for guiding the tool into alignment into the check valve **100**. The tip **606** may be the same as or similar to the tip **402** of the catheter **400**. The biluminal connector **600** may further have an opening **608** in the side wall of the biluminal connector **600**. The opening **608** may allow entry and/or exiting of the inflation and/or deflation fluid.

(55) The immobilization of the balloon with the grasping of the string can be performed with the grasping tool, as previously described. The grasping tool may be alone in the gastroscope channel and the inflation may be performed in a second channel, separated from the gastroscope channel. The biluminal connector **600** may have two lumens, as best depicted in FIGS. **14B** and **14C**. A first, central lumen **612** may extend from the proximal end **602** (see FIG. **14A**) to the distal end **604** (see FIG. **14E**). Once coupled to the gastroscope, the first lumen **612** may allow entry of the grasping tool **500** through the gastroscope channel and through the first lumen **612** to allow for grasping, alignment, and impaction of the check valve **100** as previously described. The first lumen **612** may be wholly separated from a second lumen **614** (FIGS. **14B**, **14C**). The second lumen **614** may be an annular lumen extending circumferentially around the outer surface of the first lumen **612** such that the first lumen **612** and the second lumen **614** are concentric. The second lumen **614** may be located distally to the gastroscope **800**. The second lumen **614** may be open distal to the gastroscope channel and allow entry from an outer side channel parallel to the gastroscope **800**. The entry may be located through opening **608** such that a flexible tube may be coupled to opening **608** to allow for bi-directional flow (inflation and/or deflation) through the flexible tube, second

lumen **614**, and check valve **100** (not depicted). The external channel provided in the flexible tube may run parallel to the gastroscope **800** and then may turn in to the distal end of the gastroscope **800** and enter the second lumen of the biluminal connector **600**.

(56) Referring to FIG. **12**, the biluminal connector **600** may have a silicone sleeve **700** to secure the biluminal connector **600** to the gastroscope **800**. The sleeve **700** may have an opening **708** which aligns with the opening **608** of the biluminal connector **600**. The biluminal connector **600** may be stabilized on the distal end of the gastroscope **800** (FIG. **13**) with the sleeve **700**. The biluminal connector **600** may further have a plug **610**. The plug **610** may have a rounded or atraumatic distal tip that may be placed into the distal opening of the biluminal connector **600**. The plug **610** may be pushed out by the grasping tool at any time. The plug **610** may be silicone or other flexible rubbery material or can be made of a material that melts at body temperature such as those used with medication capsules.

(57) Referring to FIG. **13**, the sleeve **700** may fit snugly around the distal end of the gastroscope **800**. The inflation tube may be external to the gastroscope and enter through the opening **708**. The biluminal connector **600** may have the proximal end **602** sitting within the distal channel of the gastroscope **800**. The distal tip **606** may enter the check valve **100** (FIG. **3**) and move the plunger head, the plunger arm, and the plunger tail distally. Near the distal end of the biluminal connector **600**, the body widens with its tapering profile to impact within the check valve **100**. As may be appreciated, the biluminal connector may prevent contamination of inflation fluid with gastric fluid and may allow the gastroscope channel to utilize its suctioning function during the procedure.

(58) FIGS. **14A-14D** depict cross-sectional views of the biluminal connector **600** taken along the lines A, B, C, and D of FIG. **1**. FIG. **14E** depicts a cross-sectional view of the biluminal connector **600** at the distal end. FIG. **14A** shows the first lumen **612** which sits within the gastroscope channel and allows entry of the grasping tool. FIG. **14B** shows the opening **608** through which the external inflation channel enters into the second lumen **614** which is an outer concentric lumen to the first lumen **612**. FIG. **14C** shows the concentric biluminal with supporting walls **616**. FIG. **14B** shows the first lumen **612** through which the tool travels and grasps the string **200** (FIG. **3**).

(59) The biluminal connector **600** may be made from silicone or a firmer material, such as Teflon or other synthetic material. The biluminal connector **600** may have a wall thickness of 0.2 to 5 mm with a central lumen of 0.8 to 2.20 mm that accommodates a grasping tool with an external diameter of 0.7 to 3.0 mm. The biluminal connector **600** may be 6-15 mm long and 3-12 mm wide.

(60) Referring to FIG. **15**, an alternative approach to accessing the balloon valve within the stomach is to separate the immobilization of the balloon from the accession of the valve. One method of immobilization is to place handles on the balloon (not depicted) that are grasped by the scope. Following this immobilization step, the valve (check valve, duck valve, or any type of valve) is accessed with a catheter that enters from the lumen of the endoscope into the valve and opens the valve, as previously described. Upon withdrawal of the catheter the valve closes. Following closure of the valve, the immobilization of the balloon is discontinued. As an example of the handles on the balloon one could place two loops of string (nylon for example) at the 3 o'clock and 9 o'clock positions of the valve housing. The two loops may be grasped by two grasping forceps **900** that exit at the 3 o'clock and 9 o'clock positions in a sleeve **902** that surrounds the endoscope **904**. The two forceps **900** grasp the two loops (not depicted) and pull them toward the endoscope tip and immobilize the balloon. The sleeve **902** may be a silicone sleeve with a lumen for one or more grasping forceps **900**.

(61) FIGS. **17-21** depict a check valve assembly similar to the check valve **100**. As may be appreciated, the exact dimensions and relationships of the valve and plunger may be altered to achieve optimized alignment, immobilization, and actuation of the check valve. For example, as depicted in the embodiment of FIGS. **17-21**, the spring may have more or fewer coils. The easy grasp loop of the string may be larger or smaller than depicted in FIG. **3**. The string may be formed from multiple strings tied or fastened together. The balloon wall may have a longer stem or a

differently contoured outer wall. Other modifications to the balloon, check valve, and catheter are within the scope of this disclosure.

(62) Referring to FIGS. **22-24**, operation of the check valve **100** or check valve of FIGS. **17-21** may be appreciated. For clarity, the catheter **400** is omitted from FIGS. **22-24**, but one may appreciate that the catheter **400** would be aligned with the opening to impact the check valve to effectuate opening of the check valve. In the open position of FIG. **22**, the plunger including plunger head **118**, plunger arm **120**, and plunger tail **122** may be seen extending out of the housing **116**. In this position, the o-ring **128** is not seated in the opening of the housing **116**. The force of the catheter (not depicted) has moved the plunger (including plunger head **118**, plunger arm **120**, and plunger head **122**) toward the interior of the balloon, compressing the spring **124** between the plunger head **118** and the block **130**, and moving the o-ring **128** off of the seat of the opening into housing **116**. Thus, fluid is allowed to flow between the interior of the valve and the interior of the balloon. The fluid may be seen to flow around the string **200**, around the plunger head **118**, plunger arm **120**, and plunger tail **122**, and through the valve opening past the o-ring **128**. It may be appreciated that during inflation, fluid may be introduced into the housing **116** through the endoscope and through catheter (not depicted) to inflate the balloon. During deflation, fluid may be removed in the reverse direction to remove fluid from the interior of the balloon.

(63) With reference to FIG. **23**, the check valve **100** is still in the open position, but is depicted rotated 90 degrees from the position of FIG. **22**. Fluid flow in FIG. **23** flows around the spring **124**, plunger **118**, **120**, **122**, and through the distal end opening of the check valve **100**. Referring now to FIG. **24**, the check valve **100** is depicted in the closed position. As may be appreciated, fluid may be located on the left side of the o-ring **128** (inside the balloon) and on the right side of the o-ring **128** (in the valve housing and in the stomach), however flow between these two areas is not permitted. The spring **124** is shown in the normally extended position, forcing the plunger head **118** toward the right and thus maintaining the o-ring **128** in the closed position.

(64) Referring to FIGS. **25-30**, another valve feature is shown and described. The valve may be similar as the valve previously described. The valve of FIGS. **25-30** may be provided in similar environments such as in a gastrointestinal balloon and/or other remote locations. As previously described, the valve may be in an unstable, remote location (e.g., the gastrointestinal balloon within a patient's stomach). The valve may include a docking station, as previously described, to allow for the immobilization, accessing, and opening and closing of the valve. The valve of FIGS. **25-30** may be accessed with a catheter, as previously described. The catheter may be deployed into the docking end or docking station of the valve to allow for access to the interior of the balloon to inflate and/or deflate the balloon. As in previous embodiments, the catheter may track centrally over an attachment (e.g., a string) to allow for the catheter to access the docking end/docking station and subsequently open and/or close the valve.

(65) FIGS. **25-28** show a valve according to another embodiment. The valve **920** may be a valve assembly. The valve **920** and the balloon **924** may have a centerline **26-26**. The valve **920** may include one or more valves **922**. The one or more valves **922** may be arranged in series. That is, a centerline that is coaxial along the axis **26-26** may run through a center of each of the one or more valves **922** such that the one or more valves **922** are coaxial. Although two valves **922a**, **922b** are shown, more or fewer valves may be provided. One valve **922** may be sufficient to allow for the access and docking of the tool within the valve. However, two or more valves **922** may be provided for redundancy, that is, to ensure proper operation of the docking station and back-up sealing between the interior and exterior of a balloon **924**. The valves **922** may be formed as cylindrical discs with an opening therethrough. The opening may be a central opening extending from a top surface to a bottom surface of the valve **922**. The opening may be tapered or conical to facilitate entry of a tool or may be a pinhole opening through the length of the valve that tightly seals around the attachment **928** or tool. The valve **920** may be considered a docking station. That is, the valve **920** may allow for the guiding, alignment, and entering of a tool into a remote area, such as, for

example, a gastrointestinal balloon. As discussed previously, the remote location may be mobile, for example, the floating gastrointestinal balloon within the stomach of a patient. The mobility of the remote location may make the valve **920** difficult or impossible to access. Such mobility may require, in conventional valves, the removal of the valve from the remote location to provide a stable environment for accessing the valve. The valve **920** may provide a docking station that allows for the immobilization and subsequent inflation/deflation of the balloon while the balloon remains in the remote, mobile or floating location. That is, the valve **920** allows for the docking, immobilization, and opening and/or closing while the gastrointestinal balloon remains in the patient's stomach.

(66) The valve **920** maybe coupled to the balloon **924**, such as previously described. The balloon **924** may have an opening **926** for receiving the valve **920**. As shown and described with respect to previous embodiments, the opening **926** may be located within the balloon **924**. That is, the opening **926** may not extend past an exterior surface **924a** of the balloon **924**. The opening **926** may have an end **926a** that extends downward from the exterior surface **924a** and into the interior of the balloon. The balloon **924** is shown truncated for clarity in FIGS. **25** and **26**. However, the balloon **924** may extend in a manner that creates a cavity that is accessible through the valve **920**. The opening **926** may be substantially cylindrical such that the opening **926** is a cylindrical bore surrounded by a wall of the balloon **924**. The opening **926** may house the one or more valves **922** and the one or more spacers.

(67) The valve **920** may include an attachment **928**, such as an elongated member, such as a string that may be have a connector **930**. The connector **930** may be an “easy grasp” loop that exits into the exterior space (e.g., the space surrounding the balloon) of the balloon **924**. The attachment **928** with easy grasp loop or connector **930** may be formed of nylon. The attachment **928** may be an elongated member, a bungee, a stretchy member, a string, a cable, a cord, other elongated device, or anything in accordance with the principles of invention capable of coupling to the valve **920** and being grasped by a tool, such as described with respect to previous embodiments. For ease of description, the attachment **928** is described as a string and the connector **930** is described as a loop, however other modes are contemplated. The string or attachment **928** may be a single string having opposing distal ends **928a**, **928b** (FIG. **27**), each coupled to a block **932**. The block **932** may be a beam or bar to which the distal ends **928a**, **928b** are fixed or coupled. Alternatively, the block **932** may be the distal ends **928a**, **928b** fixed together such that they may not pass through an opening in the spacer **934**. The block **932** may form a T-shape (for example, see FIG. **27**) such that the string or attachment **928** may be prohibited or preventing from detaching or uncoupling from the valve **920**. Although the block **932** is shown in a left to right configuration in FIG. **25**, the block **932** may be orientated along any axis of the valve **920** and/or balloon **924** so long as the block **932** prevents removal of the attachment **928** from the valve **920**. The loop or connector **930** is shown in an enlarged circular shape in FIGS. **25-28**, however the loop or connector **930** may take a smaller shape, such as described with respect to FIG. **3**.

(68) As mentioned, the valve **920** may include one or more valves **922**. The valve **920** may include one or more spacers **934**, **936**, **938**. A first spacer **934** may be located at a first end of the valve **920**. A second spacer **936** may be located on the first spacer **934**, as shown in FIG. **26**. A first valve **922a** may be located on the first spacer **934**. A third spacer **938** may be located on the first valve **922a**. A second valve **922b** may be located on the third spacer **938**. Although shown as separate components one or more of the one or more valves **922** and/or the spacers **934**, **936**, **938** may be formed integrally and/or coupled together. That is, for example, the first spacer **934** and the second spacer **936** may be formed as a single, integral component.

(69) Each of the one or more valves **922** and the one or more spacers **934**, **936**, **938** may have an opening extending therethrough. For example, the first valve **922a** and the second valve **922** may have openings **940a**, **940b**, respectively, extending through the valves. The one or more spacers **934**, **936**, **938** may have an opening **934a**, **936a**, **938a**, respectively, extending through the spacers.

The openings **940a**, **940b**, **934a**, **936a**, **938a** may be coaxial and/or aligned. The attachment **928** may extend through the openings **940a**, **940b**, **934a**, **936a**, **938a** and be coupled to or integral with the block **932**. That is, the first distal end **928a** and the second **928b** may be coupled to, integral with, or otherwise attached to the block **932**. The attachment **928** may be extended through the openings of the one or more valves and the one or more spacers such that the loop or connector **930** extends out of the opening **926** and into the exterior space around the balloon **924**.

(70) The opening **934a** may be sized such that the attachment **928** may extend therethrough and also such that the block **932** is prohibited from extending through the opening **934a**. In this manner, the block **932** may anchor or hold the attachment **928** to the valve **920**. The openings **936a** and/or **938a** may be formed of any size that is sufficient to allow passage of the attachment **928** and the tool, to be described later.

(71) The openings **940a**, **940b** of the one or more valves **922** may be formed of a size just large enough to accommodate the attachment **928**. That is, the attachment **928** may be bounded on all sides by walls of the valve **922** such that no air or other fluid may flow through the one or more valves **922**. The one or more valves **922** may be formed of a flexible material such that, upon insertion of a tool, the openings **940a**, **940b** may be enlarged to allow passage of the tool. The one or more valves **922** may be formed of a solid silicone. The one or more valves **922** may be a self-sealing valve. The self-sealing valve may be formed of a self-sealing material, such as, for example, silicone. The one or more valves **922** may be formed of a flexible material, such as solid silicone, with a small hole or opening at or around the center of the valve. That is, the one or more valves **922** may be formed of a material that is allowed to be opened or expanded from a first state (closed) to a second state (open) and back to the first state (closed). The one or more valves **922** may be formed of a material that allows for repetitive opening and closing. The openings **940a**, **940b** may be in the normally closed position, the position where the attachment **928** extends therethrough but no fluid is permitted to pass through. The openings **940a**, **940b** may be transitioned from the normally closed position to the open position by advancing a tool through the opening **940a**, **940b** to expand, distort or otherwise flex the openings **940a**, **940b** to accommodate the tool.

(72) As mentioned, the valves **922** may be formed of a flexible material with a tiny, pinhole sized hole extending therethrough. The pinhole may be the opening **940** and may be sized to accommodate the attachment **928** in the closed position but also prevent flow therethrough. In the open position, the pinhole may be able to accommodate both the attachment **928** and the tool. The opening **940** may tightly surround the attachment **928** in the closed position and may tightly surround the tool in the open position. The opening may be a pinhole opening that runs the length of the flexible (e.g., silicone) valve **922**. The valve **922** may be a loose duck-valve or duck-valve like valve.

(73) Thus, in a first state, the openings **940a**, **940b** may have a first size sufficient to allow passage of the attachment **928** but not sufficient to allow passage of fluids and/or passage of a tool with a diameter larger than the attachment. In a second state, the openings **940a**, **940b** may have a second size sufficient to allow passage of the attachment **928** and sufficient to allow passage of the tool with a diameter larger than the attachment. In the second state, a fluid may be passed through tool and thus through the one or more valves **922**. The fluid may exit through ports **942** in the first spacer **934** and allowed to flow into the interior of the balloon **924**. Although 8 ports **942** are shown in FIG. 28, more or fewer may be provided. As many ports **942** as possible may be provided in the spacer **934** such that the flow through the spacer **934** from the catheter may be as great as possible without reducing the integrity of the valve **920** and/or without allowing the attachment **928** to disengage from the valve **920**. Additionally, the ports **942** may take alternative shapes or sizes. For example, two c-shaped ports may be provided or a single, annular or donut shape port may be provided.

(74) In an embodiment where a single valve **922** is present, the spacer **938** and/or the spacer **936**

may be omitted. In an embodiment where three valves **922** are present, an additional spacer may be located between the second valve **922b** and a third valve (not depicted) about the second valve **922b**. In the case where three or more valves **922** are present, the opening **926** and the associated walls of the balloon **924** may be extended further into the interior of the balloon **924** such that the entirety of the valve **920**, including all valves **922** and all spacers, are located below the surface **924a** of the balloon **924**.

(75) In an embodiment, the valve **920** may include a second block **946**, as shown in FIGS. **29** and **30**. The second block **946** may assist in preventing removal of the attachment **928** from the valve **920**. The second block **946** may be substantially spherical. Alternatively, the second block **946** may be T-shaped like the block **932**. The block **932** and the second block **946** may hold the attachment **928** in the opening **934a** and prevent removal therefrom. The second block **946** may prevent the attachment **928** from being pushed into the valve **920** and or the interior of the balloon **924**. The valve **920** and balloon **924** may have a centerline **29-29**. The assembly may be aligned along the axis **29-29** similar to the assembly of FIGS. **25-28** being aligned on the axis **26-26**.

(76) The valve **920** of FIGS. **25-28** and/or FIGS. **29-30** may be overmolded such that the valve **920** takes the appearance or may be considered a single, integral part. The valve **920** may then be inserted into the balloon **924** as a single component. The second valve **922b** may be redundant of the first valve **922a** and may allow for confirmation and/or validation of closing and opening of the valve.

(77) The valves may be separated by one or more spacers. One or more of the spacers may be optional. One or more spacers may be omitted. That is, the valves may be separated by one or more spacers, may not be separated by one or more spacers, or combinations thereof. The valves may be next to each other, together in sequence, next or near in line or sequence, adjacent, or contiguous. The valves may abut one another. That is, the bottom surface of one valve may touch or rest on the top surface of a subsequent valve. Although described with two spacers, more or fewer may be provided. Although described with two valves, more or fewer may be provided.

(78) The valve **920** may operate in a manner similar to the method described with respect to FIG. **9**. During operation, an endoscope may be extended into a patient's stomach. A catheter **400** (FIG. **9**) with a grasping tool **500** may be inserted therethrough. The grasping tool **500**, such as grasping forceps used in endoscopic procedures may be inserted into the lumen **404** of the catheter **400** through the homeostatic valve **408** (FIG. **8**). During the procedure, the balloon, valve, catheter, and grasping tool may all be located within the patient's stomach. The forceps or similar grasping tool **500** may grasp the string or attachment **928** by the easy grasp loop or connector **930** and pull the valve **920** toward the catheter **400**. At the same time, the catheter **400** may be advanced toward the valve **920**. This may easily guide the grasping tool **500** to the target or docking end, the opening **940b** of the second valve **922b**. The catheter **400** then enters the opening **940b**, passes through the openings **938a**, **940a**, and **936a**. The tip **402** (FIG. **9**) of the catheter **400** may stop or be located in a space **944** of the spacer **934**. The tip **402** may be offset from the inner surface of the spacer **934** such that fluid may easily flow from the lumen **404** of the catheter out of the ports **942** in the spacer **934**. Such alignment of the catheter **400** with the check valve **100** allows for both immobilization of the valve **920** and thus the balloon **924** and entry into the valve **920** and balloon **924**.

(79) While fixating the catheter **400** and gastroscope with one hand, the grasping tool **500** (now grasping string or attachment **928**) within the catheter **400** is pulled back with the other hand. This motion causes the tip **402** to extend through the opening **940b**. Continued pulling of the string or attachment **928** pushes the catheter tip **402** into the remaining openings and finally into the space **944**. Alternatively, once the catheter **400** is aligned with the docking end, the catheter **400** and thus the catheter tip **402** may be pushed into the openings **940b** and then continue to be pushed through opening **940a** and into the space **944**. The opening **940b** and/or the opening **940a** may be tapered to assist and ease guiding of the catheter **400** into the valve **920**. So long as the catheter **400** extends through the valve **920**, fluid may be allowed to flow into and/or out of (e.g. bi-directional flow) the

balloon **924**. To close the valve **920**, the catheter **400** may be removed, thus returning the openings **940a**, **940b** back to the first state.

(80) After inflation or deflation is completed, the catheter **400** may be removed from the docking station or entry to the valve **920**. The tool **500**, catheter **400**, and endoscope may be removed from the patient's stomach and from the patient.

(81) During the passage of the catheter **400** through the openings of the valves **922** and spacers, the catheter **400** may track over the attachment **928** which also extends through the openings of the valves **922** and spacers. That is, in the position where the valve is open and the catheter is in place for inflation and/or deflation of the balloon **924**, both the catheter **400** and the attachment **928** may extend through the openings **904a**, **940b**, **938a**, **936a**, and **944**. The openings **938a**, **936a**, and **944** may be sized to accommodate the catheter **400**. The openings **940a**, **940b** may be formed in a self-sealing material such that the openings are permitted to expand to a size accommodating the catheter **400** and then seal or contract to a position prohibiting entry of the catheter **400**. In the position where the valve is closed and the catheter is removed, the attachment **928** may extend through the openings **904a**, **940b**, **938a**, **936a**, and **944**, but fluid flow through the valve **920** and specifically valves **922** may be prohibited.

(82) Although the catheter **400** is described as impacting the valve **920**, the catheter may be forced into the valve in other manners. For example, the catheter **400** may be secured to the opening **940a** with a turning motion. That is, the catheter **400** may have a tip **402** which may be screwed or threaded into the valve opening **940a**. One may appreciate that the inner surface of opening **940a** and the outer surface of the catheter tip **402** may have complementary threads or other surfaces to allow the rotational and longitudinal movement.

(83) Accordingly, any of the aforementioned valves and/or assemblies may allow for tracking of a tool, such as a catheter, over the attachment to provide guidance, alignment, and entry of the tool into the valve. That is, the attachment may be grasped by a first tool, and a second tool, such as a catheter, may be guided or tracked over the attachment to allow entry of the second tool into the opening of the balloon, thus actuating or opening the valve located in the opening of the balloon. In this manner, the attachment may operate as a guide or tracking member to allow for access and opening of the valve. The attachment may extend through the center of the valve or valves. The catheter may guide centrally over the attachment such that the catheter enters centrally into the opening of the balloon and/or the valve.

(84) The valve of the present disclosure may be a valve having an attachment (e.g., a string) extending centrally therethrough. That is, the valve may have the attachment running or extending through the venter of the valve. The valve may be two contiguous valves with an attachment or string that runs or extends through their center and exits the balloon. The attachment or string may enable a user to grasp the attachment or string and use the attachment or string as a tract to guide the catheter or other tool directly into the valve. The central string may facilitate entering the valve and changing (e.g., inflating or deflating) the balloon volume.

(85) As may be appreciated from the foregoing disclosure, the check valve of the disclosure may be used in a gastrointestinal balloon. The disclosure further includes a device, system, and method to immobilize the balloon within the body while at the same time providing access to the valve in such a way that the volume can be adjusted. The system can include the balloon and check valve, the catheter, and tool that is configured for grasping onto part of the valve to hold it in place. The disclosure may also include the method of performing this maneuver.

(86) Additionally, the valve may be a check valve, although other valve types (e.g. ball valves, sleeve valves, etc.) may be provided. The check valve may include a spring inside that maintains the valve in a closed position and the spring is compressed to open the valve.

(87) The check valve and balloon may be suitably immobilized while a force is applied to compress the spring. The balloon may be used in the stomach, but may be used for any inflatable, implanted device. The check valve may include a portion that may be grasped by a tool inserted to

inflate/deflate the balloon. In some embodiments, this is a loop that extends out from the valve and is grasped by a grasping tool. The loop may be pulled (away from the balloon) by the catheter/device to immobilize the balloon. Grasping forceps, inserted through the catheter, may be used to pull the loop. After the catheter engages with the valve, the loop can optionally be released with the catheter remains engaged to inflate/deflate the balloon.

(88) Additionally, as described, the check valve may be used in systems, methods, and environments other than a gastrointestinal balloon. The check valve in may have a string, or other extension, extending from the check valve. The string/extension may be grabbed by a tool inserted into the flow path to access the check valve.

(89) Although the foregoing description is directed to the preferred embodiments of the invention, it is noted that other variations and modifications will be apparent to those skilled in the art, and may be made without departing from the spirit or scope of the invention. Moreover, features described in connection with one embodiment of the invention may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A valve assembly for a gastrointestinal balloon, the valve assembly comprising: a valve formed of a self-sealing material, the valve having a first end, and a second end, and a valve opening extending from the first end to the second end; a spacer located at the first end of the valve, the spacer having a first spacer opening and at least one second spacer opening; and an attachment configured to extend through the first spacer opening and the valve opening, the attachment having an anchor configured to be located within the gastrointestinal balloon and configured to prevent removal of the attachment from the first spacer opening and the valve opening, wherein, when in an open condition, the valve is configured to allow fluid flow through the at least one second spacer opening and, in a closed condition, the valve is configured to prevent fluid flow through the at least one second spacer opening.
2. The valve assembly of claim 1, wherein the valve is a first valve, the valve assembly further comprising a second valve located in series with the first valve, the second valve formed of a self-sealing material, wherein the attachment extends through the second valve.
3. The valve assembly of claim 2, wherein the spacer is a first spacer, the valve assembly further comprising a second spacer located in series and between the first valve and the second valve.
4. The valve assembly of claim 2, wherein the valve opening through the first valve is a first opening through the first valve and further comprising a second opening through the second valve, further wherein the first opening through the first valve and the second opening through the second valve are aligned.
5. The valve assembly of claim 4, wherein the first opening through the first valve and the second opening through the second valve each have a first state preventing fluid flow therethrough and a second state allowing passage of a tool.
6. The valve assembly of claim 5, wherein, in the first state, the first opening through the first valve and the second opening through the second valve are closed around the attachment.
7. The valve assembly of claim 1, wherein the spacer is configured to separate the valve from an interior of the gastrointestinal balloon.
8. The valve assembly of claim 1, wherein the self-sealing material is silicone.
9. The valve assembly of claim 1, wherein the valve is formed as a cylindrical disc with the valve opening therethrough, the valve opening configured to receive the attachment.
10. The valve assembly of claim 1, further comprising an open position and a closed position, wherein the valve is configured to move to the open position in a stomach.
11. The valve assembly of claim 10, wherein the open position permits passage of a tool and the closed position prohibits flow of fluid.

12. The valve assembly of claim 1, wherein the anchor is a block configured to prevent removal of the attachment from the valve and the spacer.
 13. The valve assembly of claim 1, wherein the attachment extends through a center of the valve.
 14. A method of accessing a gastrointestinal balloon, the method comprising: grasping an attachment extending through a self-sealing valve; tracking a tool over the attachment and toward the self-sealing valve; pulling the self-sealing valve toward the tool simultaneously with the tracking; pushing the tool through an opening in the self-sealing valve, still tracking over the attachment, thus opening the self-sealing valve; and inflating or deflating the gastrointestinal balloon through the tool.
 15. The method of claim 14, wherein tracking the tool over the attachment comprises passing the attachment through a lumen of the tool.
 16. The method of claim 14, wherein pushing the tool through an opening in the self-sealing valve comprises expanding the opening by entering a distal end of the tool into the opening.
 17. The method of claim 14, wherein the self-sealing valve is a series of two self-sealing valves formed of silicone.
 18. The method of claim 14, further comprising removing the tool from the self-sealing valve thus closing the self-sealing valve.
 19. A system for accessing an interior of a gastrointestinal balloon, the system comprising: a gastrointestinal balloon; two valves having coaxially aligned, tapered openings; and a string running through a center of each of the two valves and exiting the gastrointestinal balloon; and an anchor configured to permanently hold the string in the two valves, the anchor located within the gastrointestinal balloon, wherein the string is configured to be grasped by a user to track a tool over the string and directly through the two valves and into the interior of the gastrointestinal balloon and wherein the anchor is fixed with respect to the two valves.
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